NORYL™ Resin PX9406P - Asia

Polyphenylene Ether + PS

SABIC

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Technical Data

Product Description

NORYL PX9406P resin is a non-reinforced blend of polyphenylene ether (PPE) + polystyrene (PS). This injection moldable grade contains non-brominated, non-chlorinated flame retardant and carries a UL94 flame rating of 5VA at 2.5mm and V0 at 0.75mm. NORYL PX9406P is VDE certified IEC60335 and offers strong electrical performance at thin wall, high heat resistance, Low Warpage, low moisture absorption, and dimensional stability. This material is an excellent candidate for unattended appliance applications such as printed circuit board (PCB) holders, terminal blocks + connectors, circuit protection / relay / switches, motor end caps, control box / panels, and battery charger / fuse holders where VDE IEC60335 is required.

General

Material Status	• Commercial: Active		
UL Yellow Card [1]	• E45587-237097 • E45587-237098 • E207780-631706 • E45587-631707 • E207780-228585		
Search for UL Yellow Card	• SABIC • NORYL™ Resin		
Availability	• Asia Pacific		
Features	• Amorphous • Bromine Free • Chlorine Free • Flame Retardant	• Good Dimensional Stability • Halogen Free • Hydrolytically Stable • Low Moisture Absorption	• Low Shrinkage • Low Specific Gravity • Low Warpage
Uses	• Appliances • Electrical Parts	• Electrical/Electronic Applications • Electronic Displays	• Solar Applications
Processing Method	• Injection Molding		
Also Available In	• Europe	• Latin America	• North America

Physical	Nominal Value	Unit	Test Method
Density / Specific Gravity	1.10	g/cm³	ASTM D792 ISO 1183
Melt Mass-Flow Rate (MFR) (250°C/10.0 kg)	3.9	g/10 min	ASTM D1238
Melt Volume-Flow Rate (MVR) (280°C/5.0 kg)	10	cm³/10min	ISO 1133
Molding Shrinkage - Flow (3.20 mm)	0.50 to 0.70	%	Internal Method
Water Absorption (24 hr, 23°C)	0.070	%	ASTM D570
Outdoor Suitability	f1		UL 746C

Mechanical	Nominal Value	Unit	Test Method
Tensile Stress			
Yield	63.0	MPa	Internal Method
Yield	73.0	MPa	ISO 527-2/50
Tensile Strain			
Break	8.5	%	Internal Method
Break	8.0	%	ISO 527-2/50
Flexural Modulus			
--	2350	MPa	ASTM D790
-- [3]	2500	MPa	ISO 178
Flexural Strength			
--	99.0	MPa	ASTM D790
-- [3, 4]	110	MPa	ISO 178

Form No. TDS-152247-en

Document Created: Wednesday, December 15, 2021
Added to Prospector: June 2011
Last Updated: 12/2/2021

Impact	Nominal Value	Unit	Test Method
Charpy Notched Impact Strength [5] (23°C)	11	kJ/m²	ISO 179/1eA
Charpy Unnotched Impact Strength [5] (-30°C)	9.0	kJ/m²	ISO 179/1eU
Notched Izod Impact			
23°C	140	J/m	ASTM D256
-40°C [6]	8.0	kJ/m²	ISO 180/1A
-30°C [6]	6.0	kJ/m²	ISO 180/1A
-20°C [6]	9.0	kJ/m²	ISO 180/1A
-10°C [6]	10	kJ/m²	ISO 180/1A
0°C [6]	11	kJ/m²	ISO 180/1A
23°C [6]	10	kJ/m²	ISO 180/1A

Thermal	Nominal Value	Unit	Test Method
Deflection Temperature Under Load			
0.45 MPa, Unannealed, 6.40 mm	136	°C	ASTM D648
1.8 MPa, Unannealed, 6.40 mm	122	°C	ASTM D648
1.8 MPa, Unannealed, 4.00 mm, 64.0 mm Span [6]	117	°C	ISO 75-2/Af
Vicat Softening Temperature			
--	140	°C	ISO 306/B120
--	150	°C	ISO 306/A120
Ball Pressure Test [7] (125°C)	Pass		IEC 60695-10-2
CLTE			
Flow : -30 to 30°C	7.0E-5	cm/cm/°C	TMA
Flow : -40 to 40°C	8.0E-5	cm/cm/°C	ISO 11359-2
Transverse : -40 to 40°C	8.0E-5	cm/cm/°C	ISO 11359-2
RTI Elec	110	°C	UL 746B
RTI Imp	105	°C	UL 746B
RTI Str	110	°C	UL 746B

Electrical	Nominal Value	Unit	Test Method
Surface Resistivity	1.0E+16	ohms	ASTM D257
Dielectric Constant			ASTM D150
60 Hz	2.80		
50 kHz	2.80		
Arc Resistance [8]	PLC 6		ASTM D495
Comparative Tracking Index (CTI)	PLC 2		UL 746A
High Amp Arc Ignition (HAI) (> 0.75 mm)	PLC 0		UL 746A
High Voltage Arc Resistance to Ignition (HVAR)	PLC 4		UL 746A
Hot-wire Ignition (HWI) (> 0.75 mm)	PLC 0		UL 746A

Flammability	Nominal Value	Unit	Test Method
Flame Rating			UL 94
> 0.40 mm	V-2		
> 0.75 mm	V-0		
> 2.0 mm	5VB		
> 2.5 mm	5VA		
Glow Wire Flammability Index			IEC 60695-2-12
0.75 mm	960	°C	
1.0 mm	960	°C	
1.5 mm	960	°C	
3.0 mm	960	°C	

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Flammability	Nominal Value Unit	Test Method
Glow Wire Ignition Temperature		IEC 60695-2-13
0.75 mm	750 °C	
1.0 mm	775 °C	
1.5 mm	775 °C	
3.0 mm	775 °C	

Fill Analysis	Nominal Value Unit	Test Method
Melt Viscosity (280°C, 1500 sec^-1)	278 Pa·s	ISO 11443

Injection	Nominal Value Unit
Drying Temperature	105 to 110 °C
Drying Time	3.0 to 4.0 hr
Suggested Max Moisture	0.020 %
Suggested Shot Size	30 to 70 %
Rear Temperature	245 to 295 °C
Middle Temperature	255 to 300 °C
Front Temperature	265 to 305 °C
Nozzle Temperature	275 to 305 °C
Processing (Melt) Temp	275 to 305 °C
Mold Temperature	70 to 100 °C
Back Pressure	0.300 to 0.700 MPa
Screw Speed	20 to 100 rpm
Vent Depth	0.038 to 0.051 mm

Injection Notes

- Drying Time (Cumulative): 8 hr

Notes

[1] A UL Yellow Card contains UL-verified flammability and electrical characteristics. UL Prospector continually works to link Yellow Cards to individual plastic materials in Prospector, however this list may not include all of the appropriate links. It is important that you verify the association between these Yellow Cards and the plastic material found in Prospector. For a complete listing of Yellow Cards, visit the UL Yellow Card Search.

[2] Typical properties: these are not to be construed as specifications.

[3] 2.0 mm/min

[4] at Yield

[5] 80*10*4 sp=62mm

[6] 80*10*4 mm

[7] Approximate Maximum

[8] Tungsten Electrode

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Where to Buy

Supplier

SABIC
Web: http://www.sabic.com/

Distributor

3Polymer (Guangzhou) Chemical Technology Co., Ltd.
Telephone: +86-20-3466-7988
Web: http://3polymer.com
Availability: China

Form No. TDS-152247-en

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Added to Prospector: June 2011
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